

HyunSeo Jung

Department of Mechanical Engineering
Pohang University of Science and Technology
39 Jigok-ro, Nam-gu,
Pohang 37666, South Korea

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Email: hsjung02 [at] postech [dot] ac [dot] kr
Homepage: <http://hsjung02.github.io>
Youtube: <http://youtube.com/@hsjung02/>

RESEARCH INTERESTS

Robotics, Automation, Control theory, Artificial Intelligence

EDUCATION

Pohang University of Science and Technology (POSTECH) <i>Ph.D. Student in Mechanical Engineering</i>	Pohang, South Korea Feb. 2026 – present
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Pohang University of Science and Technology (POSTECH) <i>Master of Science in Mechanical Engineering</i>	Pohang, South Korea Feb. 2024 – Feb. 2026
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- **Thesis:** On the Stability and Performance of Discrete-time PID Tracking Controllers for Lagrangian Systems
- **Advisor:** Prof. Keehoon Kim
- **Committee:** Prof. Wan Kyun Chung, Prof. Youngjin Choi, and Prof. Keehoon Kim

Pohang University of Science and Technology (POSTECH) <i>Bachelor of Science in Mechanical Engineering and Electrical Engineering; GPA: 4.02/4.30</i>	Pohang, South Korea Feb. 2020 – Feb. 2024
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- **Thesis:** Development of a Dynamic speed bump using emergency locking retractor mechanism
- **Thesis:** Reinforcement learning based Chip placement automation

Busan Science High School <i>High School Diploma</i>	Busan, South Korea Mar. 2018 – Jan. 2020
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PUBLICATIONS

Journal Articles

[J1] **H. Jung**, J. Kim, and K. Kim, “Learning Spatially Ambiguous Trajectories from Demonstrations via Phase-Modulated Dynamical Systems”, *IEEE Robotics and Automation Letters (RA-L)*, 2026

Conference Papers

[C1] **H. Jung** and K. Kim, “Safety Filter for Underactuated Robotic Arms with Velocity Inputs”, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026

HONORS AND AWARDS

POSTECHian Fellowship (POSTECH) (₩3,000,000 ≈ \$2,000)	2024
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Best Student Award (Dept. of M.E., POSTECH) ◦ Awarded to a <i>POSTECH M.E.</i> student who spent an excellent semester	2022
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Haesuk Cultural Scholarship (Haesuk Scholarship Foundation) (₩8,000,000 ≈ \$6,400)	2022-2023
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Undergraduate Group Research Program (POSTECH) ◦ Topic: Development of a lipreading device ◦ 3rd prize	2021
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POSTECH Programming Contest ◦ 2nd prize	2021
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POSTECH Presidential Fellowship (POSTECH) (about ₩40,000,000 ≈ \$32,000) ◦ Full ride program for undergraduate course	2020-2024
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Hanseong Fellowship (Hanseong Sonjaehan Scholarship Foundation) (₩10,000,000 ≈ \$8,000) ◦ Awarded to academically excellent students	2019
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RESEARCH EXPERIENCES

- Research Intern** (MARCH Lab, POSTECH) Jun. 2022 - Aug. 2022, Jun. 2023 - Nov. 2023
- Research on balancing control of an unmanned unicycle
- Freshman Research Intern** (Dept. of E.E, POSTECH) Sep. 2020 - Dec. 2020
- Practice PID control using an inverted pendulum and 2-dof helicopter system
- Freshman Research Intern** (Dept. of M.E, POSTECH) Sep. 2020 - Dec. 2020
- Practice PID control using a segway system

TEACHING EXPERIENCES

- Teaching Assistant** (Dept. of M.E., POSTECH) 2024-2026
- Graded tests and homeworks, and developed experimental setups for undergraduate course “Introduction to Robotics(MECH439)”
 - Developed experimental setups for undergraduate course “Freshman Research Intern(MECH199)”
- Student Mentoring Program** (Dept. of M.E., POSTECH) 2023
- Served as a tutor to teach Systems Control course
- Teaching Assistant** (K-MOOC) 2021-2022
- Developed course materials and assisted course operation for K-MOOC “Introduction to quantum physics and computing” lecture
- Seminar Host** (PoApper) 2021
- Served as a mentor for frontend seminar at PoApper, the POSTECH Web/App development group

EXTRACURRICULAR ACTIVITIES & MISCELLANEOUS

- Chief Financial Officer** (Undergraduate Association, POSTECH) 2023
- Student President** (Dept. of M.E., POSTECH) 2022
- President of POSTECH Table Tennis club** 2021-2022
- Member of POSTECH Cheerleading team** 2020-2021

SKILLS

- **Languages:** Korean(native), English(advanced - TOEIC 920, TEPS 454)
- **Hardware:** 3D printer, Lathe, Milling machine, Arduino, Raspberry pi
- **Software:** 🧩(including PyTorch and PyBullet framework), MATLAB, C/C++, CAD(Solidworks & Autodesk Inventor)